

K_Period_OHLC_Relative_Position_Tracker_V1

Functional Specification Document

Revision Note

- This revision reflects the implemented study behavior as of 2026-04-04.
- The study is now PV-first and banner-driven. Analytical subgraph plotting is no longer required.

1. Purpose

- Track interaction of lower timeframe bars (for example, 1-minute bars) relative to higher timeframe bars.
- Expose the resulting state through persistent variables for use by other studies.
- Display a compact status banner that updates only when its displayed values change.

2. Core Logic

- Evaluate only on closed bars.
- Use manual looping only: 'sc.AutoLoop = 0'.
- Do not simulate Period OHLC behavior internally.
- Treat a change in referenced HTF OHLC values as a new higher timeframe period and re-evaluate.
- Study graph region is '0'.

3. Reference Values

- 'PO = Period Open'
- 'PH = Period High'
- 'PL = Period Low'
- 'PC = Period Close'
- 'UpperInner = max(PO, PC)'
- 'LowerInner = min(PO, PC)'
- 'Mid50 = PL + 0.5 * (PH - PL)'

4. Zone Classification

- Zone is determined from the lower timeframe close relative to 'PH', 'UpperInner', 'LowerInner', and 'PL'.
- Encoded values are:
 - '10 = ABOVE_HIGH'
 - '1 = L1'
 - '2 = L2'
 - '3 = L3'
 - '-10 = BELOW_LOW'

Zone Layout

- 'L1 = High-side outer band = PH -> UpperInner'
- 'L2 = Inner body band = UpperInner -> LowerInner'
- 'L3 = Low-side outer band = LowerInner -> PL'

5. Midpoint State

- '-1 = BELOW_50'
- '0 = AT_50'
- '1 = ABOVE_50'

6. Width Calculations

- 'L1 = PH - UpperInner'
- 'L2 = UpperInner - LowerInner'
- 'L3 = LowerInner - PL'

Percent Width Display

- The banner shows the relative split of the HTF range as 'L1:L2:L3'.
- This corresponds to 'top:middle:bottom'.

7. Breach Logic

- High breach per bar: 'BarHigh > PH'
- Low breach per bar: 'BarLow < PL'

8. Per-Period Breach State

- '1 = INSIDE_ONLY'
- '2 = HIGH_BREACHED_ONLY'
- '3 = LOW_BREACHED_ONLY'
- '4 = BOTH_SIDES_BREACHED'

9. Engulfing Logic

- '1 = Bullish' if 'close > PH'
- '-1 = Bearish' if 'close < PL'
- '0 = Non-directional' otherwise

10. Reset Conditions

- Reset all per-period breach tracking state when referenced Period OHLC values change

11. Inputs

- Banner Horizontal Position
- Banner Vertical Position
- Period Open Reference
- Period High Reference
- Period Low Reference
- Period Close Reference
- Enable Debug Logging

Default Reference Subgraphs

- Open reference subgraph defaults to '0'
- High reference subgraph defaults to '1'
- Low reference subgraph defaults to '2'
- Close reference subgraph defaults to '3'

12. Banner

- Banner is drawn in chart region '0'.
- Banner updates only when any displayed value changes.
- Banner content format is:
- 'L1%:L2%:L3%:HighBreach:LowBreach:Zone:ClosePosition'

Banner Fields

- 'L1%', 'L2%', 'L3%' are percent widths of the HTF range.
- 'HighBreach' is cumulative per-period high-breach state shown as '1/0'.
- 'LowBreach' is cumulative per-period low-breach state shown as '1/0'.
- 'Zone' uses the implemented encoding '10 / 1 / 2 / 3 / -10'.
- 'ClosePosition' is the normalized HTF close location metric described below.

13. HTF Close Position Metric

- Store where the current LTF close sits inside the HTF range assuming:
 - '0 = PL'
 - '100 = PH'
- If close is above 'PH', store '101'
- If close is below 'PL', store '-101'
- Saturation is applied regardless of how far outside the range the close is.

14. Nearest-Level Distance Metrics

- Store the actual distance in ticks from the current close to the two nearest structure boundaries
- For closes above 'PH':
 - both 'd1' and 'd2' equal 'abs(close - PH)' in ticks
- For closes below 'PL':
 - both 'd1' and 'd2' equal 'abs(close - PL)' in ticks
- For closes inside the HTF range:
 - 'd1' and 'd2' are the distances to the two boundaries of the containing zone

15. Width Metrics In Ticks

- Store each level width in ticks:
 - 'L1 width ticks'
 - 'L2 width ticks'
 - 'L3 width ticks'

16. Persistent Variable Outputs

Integer PVs

- 'Int 20 = Zone Classification'
- 'Int 21 = Midpoint State'
- 'Int 22 = Current Bar High Breach'
- 'Int 23 = Current Bar Low Breach'
- 'Int 24 = Per-Period Breach State'
- 'Int 25 = Engulf State'
- 'Int 26 = Valid Period Flag'
- 'Int 27 = Period Reset Flag'

Float PVs

- 'Float 20 = L1 Width'
- 'Float 21 = L2 Width'
- 'Float 22 = L3 Width'
- 'Float 23 = Period Open'
- 'Float 24 = Period High'

- 'Float 25 = Period Low'
- 'Float 26 = Period Close'
- 'Float 27 = UpperInner'
- 'Float 28 = LowerInner'
- 'Float 29 = Mid50'
- 'Float 30 = HTF Close Position'
- 'Float 31 = Distance To Nearest Level 1 In Ticks'
- 'Float 32 = Distance To Nearest Level 2 In Ticks'
- 'Float 33 = L1 Width In Ticks'
- 'Float 34 = L2 Width In Ticks'
- 'Float 35 = L3 Width In Ticks'

17. Logging

- When 'Enable Debug Logging = Yes', log the lower timeframe bar timestamp and the cu

18. Edge Cases

- If 'PH <= PL', treat the period as invalid.
- Invalid periods zero the float metric PVs and set state PVs to neutral or invalid v